Array 27 (2025) 100487

Contents lists available at ScienceDirect

Array

journal homepage: www.sciencedirect.com/journal/array

Adaptive multimodal user interface techniques for mobile augmented reality: Frameworks, modalities and user interaction

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ARTICLE INFO

Keywords:
Mobile augmented reality
Human-computer interaction
User interface
Adaptive interaction
Multimodal interaction
User experience
User engagement

ABSTRACT

Mobile augmented reality (MAR) has revolutionised information perception and real-world interaction. The introduction of MAR has enriched users' visualisation abilities, enabling them to understand complex objects and reveal previously inaccessible aspects. However, traditional MAR systems predominantly rely on unimodal interaction techniques and the exploitation of user and contextual data has been moderate. This leads to inadequate interaction with augmented content ultimately impacting the user experience and engagement negatively. To address these shortcomings, adaptive multimodal user interface (AMUI) techniques in MAR have emerged, offering refined interaction mechanisms by leveraging multiple modalities and incorporating user and contextual information. In this paper, a systematic review of the adaptive and multimodal literature in MAR is conducted, aiming to assess the impact of AMUI techniques in MAR systems. This paper provides a comprehensive examination of the current state of AMUI-MAR systems and frameworks and identifies key research issues requiring further investigation. The findings of this paper have demonstrated that AMUI techniques have significant potential and can transform the interaction landscape of MAR, offering enhanced user experience and engagement. By leveraging multiple interaction modalities and better utilizing user and contextual information, AMUI techniques can pave the way for more natural and intuitive interactions in MAR.

1. Introduction

According to Van Kleef et al. [1], Augmented Reality (AR) overlays real-world environments with computer-generated content, creating a real-time augmented experience. AR has become a breakthrough in various sectors and is now widely adopted in industries such as construction, tourism, medicine, navigation, entertainment, education, and resource management. Initially, AR was perceived as obtrusive due to its reliance on specialized equipment like AR glasses or head-mounted displays (HMDs). However, the paradigm shift brought about by smartphones has significantly broadened the scope of AR, giving rise to Mobile Augmented Reality (MAR). The availability of advanced sensors and robust computing power in modern smartphones has contributed to the exponential development and deployment of MAR systems.

As the demand for MAR systems continues to grow, researchers are striving to design more sustainable and user-centric solutions. Earlier AR technologies, including desktop AR and HMD-based systems, were typically developed for controlled or known environments. With smartphones enabling on-the-go access, the dynamics of MAR have evolved, leading to new forms of interaction and more seamless, immersive experiences. In this context, user interfaces (UI) play a critical role. These systems are not only interactive but also generate large amounts of information that often require real-time user filtering and refinement. However, high levels of interactivity can negatively impact user experience (UX), imposing cognitive and physical demands on users [2,3].

The increasing complexity of MAR interfaces has drawn attention to the need for more intuitive interaction techniques that enhance UX. One promising area of exploration is the integration of multiple modalities such as motion, gesture, and speech to support interaction across different contexts. Interfaces in MAR systems rely on various input-output channels known as modalities [4]. A system that uses only one modality is considered unimodal [5], whereas a system combining multiple modalities is referred to as multimodal [6]. This paper investigates these interaction modalities with a focus on adaptive multimodal user interfaces in MAR.

While unimodal techniques are relatively easy to implement in MAR systems, multimodal approaches remain underexplored and are often

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https://doi.org/10.1016/j.array.2025.100487
Received 1 October 2024; Received in revised form 14 July 2025; Accepted 2 August 2025
Available online 6 August 2025
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confined to controlled setups. Furthermore, MAR applications continue to face persistent usability challenges, including small screen sizes, cognitive overload, high interaction costs, physical discomfort, information saturation, and the presentation of irrelevant content. These issues, as reported by Zhao et al. [7], Lindgren et al. [8], Aitamurto et al. [9], and De Sá & Churchill [10], present significant obstacles to human-computer interaction (HCI) and may lead to poor UX and user disengagement [11]. There is thus a pressing need to investigate adaptive multimodal techniques that can dynamically adjust to users' environments, behaviours, and needs, thereby enhancing the overall interaction experience.

Previous AR interaction models have included approaches such as 3D interaction, natural interaction, tangible interaction, spatial awareness, and multimodal interaction [5]. While most studies concentrate on interaction with virtual 3D content, this research also emphasizes interaction with the MAR system itself. Accordingly, interaction techniques in MAR can be grouped into tangible interfaces, which involve physical objects to manipulate digital elements; collaborative interfaces, which facilitate shared experiences through multiple displays; hybrid interfaces, which offer multiple interaction options such as gestures or speech in a sequential manner; and multimodal interfaces, which allow the simultaneous use of different input channels [12,13].

In real-time, interactive MAR environments where HCI is intensive, multimodal interfaces can provide more natural interactions by enabling users to engage with content using multiple inputs at once [14]. Designing such systems requires careful consideration of context-awareness, user behaviour, and device capabilities [12]. For example, a user visiting a cultural heritage (CH) site using a MAR application may use voice commands in quiet indoor spaces, while switching to gesture-based controls in noisier outdoor environments. This dynamic shift in interaction modality based on environmental context illustrates the potential of Adaptive Multimodal User Interfaces (AMUI). By leveraging real-time data to combine and adapt input modalities such as touch, gesture, speech, and motion, AMUIs aim to optimize HCI by reducing interaction costs and enhancing user experience and engagement. Fig. 1 illustrates how an AMUI system seamlessly switches between modalities in response to contextual variables like noise level, user movement, and environment.

Most existing studies on multimodal interaction have been conducted outside the MAR context, where environmental diversity and device heterogeneity are not adequately addressed. This review focuses on adaptive multimodal techniques specifically within MAR, with the goal of identifying key technical requirements, functional components, and contextual parameters that shape adaptive interfaces. The paper also aligns with contemporary research trends emphasizing intelligent, context-aware, and multimodal systems.

Mohiuddin et al. [15] proposed an edge-based architecture for reducing latency in mobile learning applications, Theodorou et al. [16] surveyed visual SLAM techniques for AR mapping and wayfinding, and Mohyuddin et al. [17] introduced a deep learning-based framework for hand gesture recognition. Building on these contributions, the present paper provides a comprehensive review of adaptive multimodal frameworks in MAR and explores how these frameworks enhance user experience through contextual interaction models. This systematic review begins by defining adaptive multimodal interaction in MAR and discussing existing frameworks. It then presents core interaction modalities and techniques, followed by an evaluation of representative MAR systems. Subsequent sections explore how adaptive techniques vary across different contexts and identify commonly used methods in the field. A synthesis of the findings is then presented through a proposed AMUI framework, before concluding with a discussion and future research directions.

2. Research methodology

Building on the gaps identified in the introduction section, this study adopts a systematic methodology to examine adaptive multimodal frameworks in MAR. While multimodal interaction has been explored in AR more broadly, limited attention has been given to its adaptive use in mobile contexts. This review investigates key components, design requirements, and contextual factors that influence human-centric interaction in MAR systems.

In this paper, a review of the adaptive and multimodal theories in MAR has been performed using the Kitchenham method [18] whereby important research questions have been formulated and narrowed down through the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) techniques where some important pieces of evidence are put forward for the analysis and evaluation of this study [19]. The investigation funnel is organised in the following steps: planning the review, conducting the review, the inclusion and exclusion criteria and finally the search phase where the review is documented.

2.1. Planning the review

Human-Computer Interaction (HCI) remains underexplored in the MAR domain, particularly in the context of Adaptive Multimodal User Interfaces (AMUI), which are gaining traction as a means to enhance user interaction. To understand the architectural components,

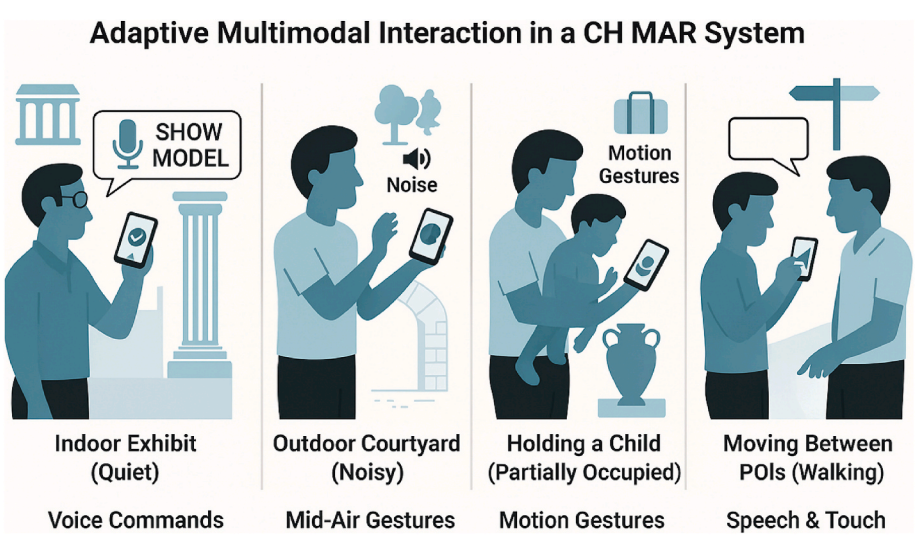

Fig. 1. Adaptive multimodal interactions in live scenario.

contextual factors, and design requirements of AMUI in MAR systems, this review assesses and classifies existing works using defined parameters. The study examines how unimodal techniques evolve into multimodal ones and investigates adaptive multimodal mechanisms across various situational contexts.

2.1.1. Research questions

The following research questions have been formulated and addressed in this paper.

RQ1 How the adaptive multimodal frameworks have improved the current state of MAR systems?

At the initial stage of this study, we will introduce the terms *adaptive* and *multimodal* in the MAR space. Moreover, recent frameworks will be investigated in this field and they will be assessed and evaluated with the current state of MAR systems.

RQ2 What are the most impactful unimodal techniques and multimodal techniques that have been employed for Adaptive Multimodal User Interface (AMUI) in MAR?

The unimodal (singular) interaction modalities are extracted from existing MAR systems and how they can be multiplexed to form multimodal techniques for AMUI in MAR. A classification will materialise in finding the trade-offs of these techniques and their usefulness in the AMUI-MAR context.

RQ3 What are the relations and interconnectedness of the various components (interaction modalities, intrinsic and extrinsic context parameters, adaptation module and overlay interface) that are involved in an AMUI framework for MAR?

The main components of the framework will be detailed. The input and output channels will be identified and described. The internal and external parameters will be determined and investigated. The adaptation module takes care of all the situational (intrinsic and extrinsic) settings and adjusts the interactivity functioning of the system accordingly. The overlay interface provides the augmented objects. The relationship between these components will be mapped and they will assist in determining and understanding the underlying components of AMUI in MAR.

RQ4 How the AMUI framework can impact the HCI (User Experience and User Engagement) in MAR?

In this study, the usefulness of the AMUI frameworks is determined in MAR. An assessment of the user experience and engagement is carried out using a qualitative analysis. The HCI factors such as *gesture, touch, gaze* and *speech* will be assessed individually.

2.2. Conducting the review

2.2.1. The searching process

According to the research questions, the review has been planned by determining the most appropriate search strategies. The steps carried out for the search process are as follows: identify the most recent and relevant adaptive multimodal MAR frameworks, the interaction modalities and output channels and the components related to the adaptive multimodal framework. The search has been carried out in Google Scholar, ACM and IEEE forums.

2.3. Inclusion and exclusion criteria

The inclusion and exclusion criteria were defined as follows:

2.3.1. Inclusion

- Most recent studies were considered. The search was limited to the last five years.
- Keywords such as mobile augmented reality, multimodal MAR systems, multimodal frameworks, adaptive user interfaces, adaptation strategies, unimodal techniques, multimodal techniques, interaction modalities, and output and input channels were queried.
- Papers related to multimodal applications, multimodal frameworks, multimodal user interfaces, multimodal techniques, multimodal fusion, adaptive user interfaces and adaptive strategies in the MAR context.

2.3.2. Exclusion

- Multimodal theories or frameworks for general or non-MAR systems.
- Studies mentioned the term “MAR” but are related to “head-worn display” or “smart-glass”.
- Studies related to virtual reality and mixed reality.
- All forms of studies which are guest editorials, unpublished working papers and master’s dissertations

Based on the above criteria, data were extracted for quality assessment.

2.4. Search

The key terms were searched in Google Scholar, ACM and IEEE forums whereby a total of 221 non-duplicated articles were retrieved. The inclusion and exclusion criteria were thus applied based on title, abstracts and full texts, and as a result, we selected only 15 articles. In the following subsections, our findings with respect to each research question are presented. Fig. 2 is a PRISMA representation of the related results obtained from the search.

3. Multimodal MAR systems

There have been relatively few works focused on developing multimodal frameworks for MAR systems. As introduced earlier, the multimodal interface incorporates multiple input modalities, enabling users to interact with both 3D objects and the MAR environment simultaneously or asynchronously. To gain a comprehensive understanding of the key components and requirements of a multimodal interface in a MAR system, it is essential to evaluate existing research and frameworks in this field. Drawing from the methodology outlined in Section 2, this chapter reviews and analyses the current state of work in this area.

3.1. Literature review

Islamic Tourism Go [20] is a location-based AR application built upon the earlier framework by Abidin et al. [12]. This application helps tourists locate Islamic Points of Interest (POI) by utilizing the users’ geographic locations. A range of sensors, including proximity, accelerometer, compass, GPS, microphone, and camera, are integrated to detect multimodal inputs. The authors employed natural interaction mechanisms, such as push-to-talk, near-to-talk, and motion gestures, to create a multimodal interface. The adaptive interface utilizes contextual data from user profiles, environmental factors, and device state information to deliver optimized interaction modalities, enhancing user engagement and experience. At the other hand, Chen et al. [4] explored gesture and speech modalities in an AR game to evaluate the effectiveness of these interaction methods within the AR environment. Using Leap Motion for gesture recognition and the Google Cloud API for speech, the authors conducted a field study with eighteen participants. The study demonstrated that utilizing both modalities concurrently resulted in greater accuracy, underscoring the potential for multimodal interaction to enhance the precision and efficiency of user input.

Srivastava & Yammiyavar [21] developed a multimodal instructional AR system tailored for students in laboratory settings. By merging Human-Computer Interaction (HCI) and machine learning techniques, the system aims to reduce cognitive load by providing contextualized instructions and information. Text-based and speech modalities were used for user interaction, demonstrating that multimodal interfaces can streamline learning processes and improve instructional outcomes in

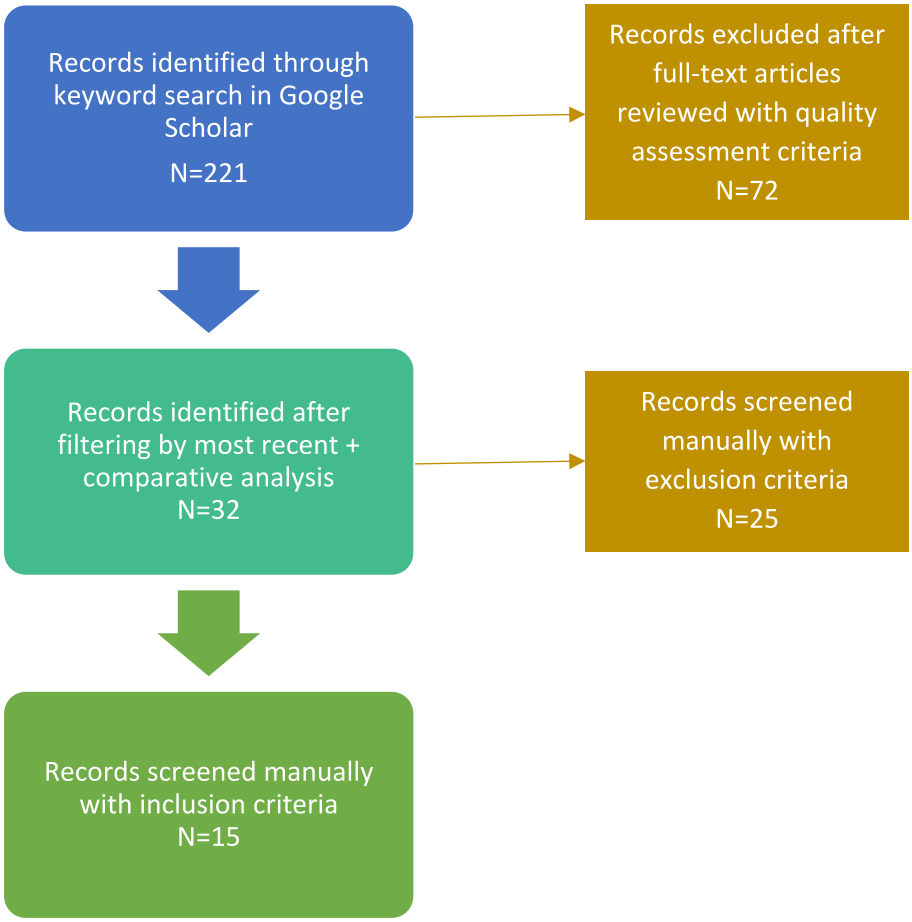

Fig. 2. PRISMA representation of the search conducted.

educational environments. Similarly, Abrami et al. [22] introduced the Vannotator framework for generating multimodal hypertexts in AR. This framework allows for the creation and annotation of texts and images linked to real objects or animations within augmented virtuality. The authors expanded the traditional understanding of multimodality by incorporating body movements and gestures in the annotation process, showcasing how multimodal interaction can extend beyond standard user-system interaction to integrate physical-world elements.

Del Amo et al. [23] proposed an information-centred framework for MAR applications in the maintenance industry. Their framework supports maintainers by providing AR-based instructions, visual content, text, and 3D models. Gesture recognition techniques were applied to track technicians' movements and assess their performance during repair activities. This study highlights the potential of multimodal MAR systems to improve task efficiency and accuracy in industrial contexts by enhancing real-time interaction. In the same vein, Hellmuth et al. [24] investigated interaction techniques for positioning 3D objects in AR environments. Four different methods for creating anchors and moving objects in a handheld AR environment were evaluated. The study's quantitative results indicated that creating an anchor on the surface and moving an anchor using finger gestures provided the most efficient and accurate interaction. This reinforces the value of combining natural gestures with object manipulation in 3D space for intuitive user interaction.

TourMar [25] is a MAR application built on the design principles outlined by Shukri et al. [26]. This application integrates multimodal data from audio, 3D images, and video, presenting users with a consolidated view for simultaneous access. The design principles emphasize context utilization, visual UI elements, and 3D interaction techniques to improve user interactivity and overall experience, highlighting the role of multimodality in enhancing MAR system usability and engagement. Likewise, Firmanda et al. [27] explored the integration of meta-user interfaces in MAR for CH education. After evaluating interface types such as non-diegetic, spatial, meta, and diegetic within the MAR space, the authors selected a meta-UI approach, similar to the diegetic interface. The system incorporated hand gestures and body tracking, and its evaluation through the PIECES framework yielded satisfying results. This study illustrates how multimodal interfaces can enrich educational experiences by blending immersive interactions with real-world objects.

Harrington [28] examined the use and impact of a MAR application within a museum diorama to enhance informal learning and visitor engagement. The design incorporated story elements, bioacoustics, and soundscapes to create a multimodal and multisensory learning experience. The findings revealed that multimodal cues significantly improved cognitive engagement and informal learning outcomes, demonstrating the potential of multimodal MAR applications to foster self-directed exploration and knowledge acquisition. AR Sightseeing, proposed by Matviienko et al. [29], examined three types of information placements in an outdoor AR environment: on-body, world, and overlay. While not explicitly focused on multimodal interactions, the evaluation of these placements provides insights into human-interaction mechanisms, such as ease of use and scene exploration. The results indicated that world placement was more convenient than overlay placement, hinting at the potential for multimodal integration in outdoor MAR applications to enhance user interaction.

Gül [30] investigated gesture-based and touch-based interactions in MAR for co-design activities. Through two user studies, the research evaluated the affordances of gesture-based and touch-based interactions in a MAR environment. While gesture-based interactions were found to

be less precise, touch-based interaction encouraged idea generation. This highlights the importance of balancing different modalities to achieve both precision and creativity in MAR applications. Schraffenberger & Heide [31] emphasized that MAR systems should bridge the virtual and physical worlds, moving beyond a purely vision-centric approach. Their work advocated for incorporating hearing, touch, smell, and taste modalities to align more closely with real-world sensory experiences. This study reframed multimodality in MAR as an avenue for creating a holistic, immersive experience by seamlessly blending virtual content with the physical environment. Building on this idea, Ismail et al. [32] designed a multimodal interface for manipulating 3D objects in AR using both hand gestures and speech. The Leap Motion technique tracked hand and finger movements, while speech commands controlled specific system actions. This study addressed common interaction challenges, such as occlusion, by enabling natural and intuitive manipulation of 3D objects. The findings underscore the utility of combining gesture and speech inputs for a more fluid and precise user experience.

Hürst & Vriens [33] compared vision, audio, and haptic modalities for finger-based interactions in MAR. The study focused on basic grab-and-release gestures, foundational to more complex interactions. Their results indicated that multimodal feedback not only improved interaction speed but was also preferred by users, demonstrating the efficiency and user appeal of multimodal input in AR environments. Extending this perspective, Lugtenberg et al. [34] explored how multimodal AR technology can alter the perception of object thickness. By manipulating auditory and tactile feedback, they demonstrated how sensory input can influence the perception of physical properties. Their findings suggest that incorporating multiple senses into AR systems can enrich user experiences by providing more nuanced and immersive interactions.

Table 1 summarizes the fifteen multimodal MAR systems analysed in this research, focusing on their application areas, input modalities, context parameters, and interaction algorithms. These systems are

Table 1
Multimodal MAR systems.

#	Multimodal MAR Systems	Application Area	Techniques			Applicable Frameworks	Limitations
			Input Modalities	*Context Parameters*	*Interaction Algorithm*		
1	Islamic Tourism Go (Abidin et al. [20]	Tourism	Gestures, Speech & Touch	Location information, User profile & Device state	Adaptive multimodal – Combining the motion, speech and touch modalities	Adaptive multimodal interface location-based AR [12]	Multimodal techniques cater only when the user is in a static location.
2	Chen et al. [4]	–	Gesture & Speech	–	Leap Motion and Google Cloud Speech API	–	Adaptive multimodal techniques have not been explored.
3	Srivastava & Yammiyavar [21]	Education	Touch & Speech	–	Intelligent Breadboard – Smart learning objects	Net MAR [35]	The modalities are limited and the interaction mechanisms are exhaustive for students.
4	Vannotator (Abrami et al. [22]	Maintenance	Touch & Gesture	Temporal Locations & Discourse Referents	Annotation of the network of interlinked units, multicodal network, teleportation and selection.	Context-Aware Application [36]	Graph drawing has not been exploited. No mid-air gesture and adaptive multimodal techniques have been explored.
5	Del Amo et al. [23]	Maintenance	Gaze & Gesture	Environmental Data and Image Tracker	Gesture interaction, 3D and 2D visualisation	ARSEM [37]	Multimodal gesture techniques could be exploited to track the interaction of real-world objects.
6	Hellmuth et al. [24]	Research	Gesture & Touch	Device movement	3D geometry manipulation – Creation and Movement methods	Adaptive multimodal interface location-based AR [12]	The speech modality has not been explored.
7	TourMar [25]	Tourism	Touch	Tourism Destination, Point of Interest, Device information	–	–	The multimodal interaction is yet to be integrated.
8	Firmanda et al., (2022)	Cultural Heritage	Gesture	Environment data, hand gestures & body motion	Meta-UI interface – 2D interface displayed to the user as an effect of the augmentation.	–	An adaptive model could be explored that leverages touch-UI and meta-UI.
9	Harrington [28]	Cultural Heritage	Touch & Gesture	Environment data & Device information	UI interface with data visualisation and story buttons	Context- Aware Application [36]	Multimodal AR is limited to the contents only.
10	AR Sightseeing [29]	Cultural Heritage	Touch	Historical cultural information	Standard UI interface with only touch modality.	–	Other modalities could have been explored to better evaluate the user experience.
11	Gül [30]	Research	Touch & Gestures	Physical Model & Geometric Primitives	Multiple interaction actions: Inspecting, Pointing, Alignment, Stacking, Arrangement, Registering, Rotating & Selecting	Context- Aware Application [36]	Mid-Air gestures and Gaze modality could have enhanced the interaction mechanism with the AR content.
12	Schraffenberger & Heide [31]	–	–	–	–	–	–
13	Ismail et al. [32]	Research	Gesture & Speech	Physical environment	6DOF with Leap Motion hand tracking and Speech recognition.	AR-Speech-Emotion-Marker (ARSEM) [37]	The occlusion effect has not been addressed in the multimodality AR system.
14	Hürst & Vriens [33]	Research	Touch & Gestures	Virtual environment	Natural feature tracking for finger tracking and Linear Resonant Actuator for Haptic feature.	Adaptive multimodal interface location-based AR [12]	Speech modality would add a further enhanced interaction. A flexible interaction model is yet to be exploited.
15	Lugtenberg et al. [34]	Research	Touch & Gestures	Physical environment	–	Context- Aware Application [36]	Research is limited to one object only,

commonly applied in fields such as heritage tourism, education, and maintenance, with CH and maintenance sectors being the most prominent. Both fields involve interacting with closely spaced real-world objects that require accurate, real-time augmentation. In CH, the goal is to deliver a natural, immersive experience with seamless interaction with AR content. In maintenance, multimodal MAR systems facilitate efficient, context-aware interventions during repairs, enhancing precision and response time.

3.2. Main findings of the AMUI-MAR systems

MAR systems have a complex multi-layered interface where the interaction cost is usually high and the amount of contents is substantial. These issues have a direct impact on the interactivity with the systems and therefore can impact the cognitive processing of the users. In this context, the multimodal system has unified one or more input modalities that would provide the users with an intuitive interface, allowing them to seamlessly manipulate and interact with the AR objects. Touch and gesture modalities have been commonly employed in the reviewed works, representing around 56 % of the total works. Touch modality is the most natural interaction mechanism in a smartphone and it is often perceived as the key point of initiator between the user and the device. In a MAR continuum, tactile augmentation is widely engaged due to its convenience and accessibility. Moreover, in constrained environments with multiple objects of interest, gesture-based interaction is recommended to enhance control over superimposed content. This allows users to explore objects in 360° and inspect details that may otherwise remain inaccessible.

Chen et al. [4] and Hellmuth et al. [24] both explore gesture-based interactions in MAR, but with different focuses. Chen et al. [4] evaluate the accuracy of combining gesture and speech modalities, using Leap Motion and Google Cloud API, showing that simultaneous use of these modalities improves interaction accuracy. In contrast, Hellmuth et al. [24] examine the effectiveness of various object placement and movement techniques using gestures in a 3D space, confirming that gesture-based interaction provides better precision when anchors are placed on surfaces. Abidin et al. [20] integrate interaction modalities based on the context, such as the user's location and device state, demonstrating the flexibility of multimodal systems to switch between modalities as needed. These studies indicate that gesture-based interactions are critical for enhancing natural user experiences in AR, particularly in applications requiring precise control, such as education, maintenance, and tourism.

Several studies have combined audio and speech modalities with visual or gesture inputs to create more immersive and hands-free interaction mechanisms. Srivastava & Yammiyavar [21] employ speech alongside text-based interaction to assist students in laboratory environments. By reducing cognitive load, the speech modality provides contextualized instructions, ensuring that users are not overwhelmed with visual information alone. Harrington [28] incorporates ambient soundscapes and bioacoustics in an AR museum diorama to create a multisensory learning experience. By adding sound elements to visual content, the system fosters engagement and enhances informal learning. This use of audio feedback highlights the potential of multimodal systems to stimulate multiple senses simultaneously, thus improving overall user interaction and engagement. These examples show how speech and audio modalities can complement visual and gesture-based interactions, particularly in applications where hands-free or natural interaction is crucial, such as in education or hands-on tasks.

Few MAR systems have adopted the adaptive multimodal interaction mechanism because of the current complexities and challenges that they have had when implementing it in a heterogeneous environment. As this segment considers the current contextual settings which are dynamic, the issue that may arise is the jitter delay from the collection of real-time data to its processing. The environmental settings, the device state, the preferred interaction modalities, or users' gestures or movements are among the list of parameters that should be considered to build the adaptive multimodal MAR system. The interaction in MAR can be separated into two axes. The interaction with the MAR system should be treated differently from that of the interaction with the augmented objects. In the scenario, where the user is interacting with the MAR system, it mainly involves navigating through the different screens and using the various functionalities and features. The interaction with the virtual (3D) contents requires manipulation with the Six Degree of Freedom (6DOF) technique, which consists of moving forward or backwards, up or down, and right or left.

Ismail et al. [32] have examined the Leap Motion technique for recognizing hand gestures in their MAR prototype. The model employed the 6DOF manipulation technique, which is divided into 3DOF for object translation and 3DOF for object rotation. The interaction with the virtual contents is claimed to be occlusion-free since the system is designed with both gestures and speech modalities. However, recent advancements in interaction techniques, such as the incorporation of 7DOF, offer even greater precision and flexibility by adding a seventh degree of freedom, typically involving an additional parameter such as scaling or more complex spatial control. Gül [30] has explored interaction mechanisms such as selecting, grabbing, pinching, rotating, scaling, arranging, and stacking 3D objects in their MAR model. In this context, the freedom of having multiple interaction mechanisms, possibly enhanced by 7DOF capabilities, has supported idea generation from the users, e.g., new touch-based techniques for interacting with the 3D elements.

The mid-air gesture technique is another prominent research area in the MAR continuum that requires a detailed investigation in the quest for multimodal interaction. This technique tracks the bare hands or finger gestures as input, enabling intuitive manipulation of 3D contents. The works of Firmanda et al. [27] and Hürst & Vriens [33] have adopted this technique to enable natural interaction with virtual objects. The findings from these works stated that this technique has provided a new sense of touch perception to the interactivity with 3D objects in MAR, making it feel more natural and immersive compared to traditional interaction modalities. Incorporating 7DOF into these systems could further enhance the user experience by allowing even more nuanced control over virtual objects.

Context-aware multimodal systems leverage environmental data (e.g., noise, light, temperature) to dynamically adapt interaction modalities, ensuring that users receive the most suitable experience based on their context. Abidin et al. [12] incorporate adaptive multimodal techniques to adjust the interaction channels based on environmental characteristics like noise or light conditions. This ensures that the system selects the most appropriate modality, such as switching to gesture-based interaction in noisy environments where voice commands may not be effective. Del Amo et al. [23] designed a framework for the maintenance industry that dynamically adapts to technicians' needs. By employing gesture and visual content, the system improves user performance in environments that demand high precision and hands-free interaction. Adaptive systems, therefore, hold promise in situations where the environment significantly impacts user experience, such as in noisy or complex workspaces.

The predominant application domains for multimodal MAR systems are CH, education, and maintenance. These fields demand seamless, intuitive interaction with virtual and physical content, often requiring multimodal feedback to enhance engagement and task efficiency. The majority of CH MAR systems, such as TourMar [25] and Firmanda et al. [27], integrate multimodal data (audio, visual, 3D content) to enhance user experience in historical sites. These systems rely on multimodal interfaces to engage users in immersive learning environments where the combination of visual and audio modalities plays a crucial role in user engagement. In the maintenance sector, Del Amo et al. [23] use gesture and visual interactions to provide technicians with hands-free access to critical information while performing complex tasks. Gesture-based input enables maintainers to interact with AR content without needing to physically manipulate their devices, allowing for

more efficient task completion in constrained environments. Despite the advancements in multimodal interactions, there are still limitations, particularly in sensory modalities beyond vision and touch. For instance, Schraffenberger & Heide [31] argue that the integration of other sensory modalities, such as smell and taste, is largely unexplored in current MAR systems. Expanding the sensory spectrum could further enhance the realism and immersive potential of MAR systems, particularly in applications such as food-related AR experiences or medical training simulations.

From the surveyed works, it is evident that certain modalities dominate the field. As noted, 56 % of the studies use touch and gesture inputs, primarily because these modalities are intuitive and accessible on mobile devices. Audio and speech modalities appear in 34 % of the papers, highlighting their role in enhancing hands-free interaction and providing complementary feedback to visual stimuli. In terms of application areas, CH and maintenance sectors lead in adopting multimodal MAR systems. Around 40 % of the analysed papers focus on CH applications, leveraging multimodal interfaces to improve user engagement with historical content. Meanwhile, 28 % of the works address maintenance tasks, where multimodal systems help users efficiently interact with technical instructions and visual content.

3.3. Impact of AMUI-MAR systems on UX and UE

The sections above have enlightened the understanding of the influential UX and UE attributes that revolves around the AMUI-MAR framework. Moreover, a thorough analysis of the AMUI-MAR is important to understand how the UX and UE aspects have evolved from the conventional MAR to the AMUI-MAR version. Table 5 provides a high-level assessment of the UX and UE impactful factors of the AMUI-MAR systems discussed in Section 3.1. The most suitable UX and UE factors from the works of Ramtohul & Khedo [38], Ramtohul & Khedo [39], Olsson [40] and Buccini & Padovani [41] to assess the systems.

From Table 5, it can be noted that most of the systems have adopted the UX and UE influential factors. The systems have been assessed in relation to the adaptive mechanism and multimodal interaction models that they have employed to improve the UX and UE. Most of these systems are already built with common metaphors and icons which are easily identifiable and recognisable by humans, thus justifying the hundred percentiles of the *familiarity* element in all the systems investigated in this research. The *familiarity* characteristic can also be associated with commonly known gestures such as tapping, swiping, swinging, dragging, pushing and pinching. Moreover, these systems comprise multi-layered interfaces where interactivity is an important aspect to consider while designing such systems. As a result, *interaction* is a key ingredient for the proper functioning of such systems. *Direct Manipulation* is an important aspect where the design of the systems is focussed on less physical effort. Islamic Tourism Go and Hürst & Vriens [33] have integrated natural interfaces such as mid-air or finger-tracking gestures to have a tangible interaction with the MAR objects. Vannotator [22] and Lugtenberg et al. [34] are among the works which have embedded the *intuitiveness* factor by using the common hand and finger gestures such as touch, move and release.

TourMar [25] and Del Amo et al. [23] have exploited the *personalisation* attribute by leveraging on the users' profile data and the contextual information. The systems can as well propose user-centric AR information to manage cognitive overload and encourage more cognitive growth. The *liveliness* factor can emotionally engage the users to perpetually use the system. The works of AR Sightseeing [29] and Hellmuth et al. [24] have integrated sophisticated modalities such as speech interaction, soundscapes and mid-air gestures to render an energetic environment. The *sense of presence* refers to tracking the users' location and superimposing the environment which makes the user feel that they are actively interacting with the real world. However, innovative interaction mechanisms such as gaze, mid-air, motion, sound and speech can enhance the *sense of presence* significantly. Harrington [28] and Lugtenberg et al. [34] are among the works which have embedded the *sense of presence* by exploiting novel interaction modalities. *Multi--interaction* is a key feature in an AMUI-MAR system, it provides several interaction modalities to interact with the systems and MAR objects. As a result, the *multi-interaction* feature should be a prerequisite in the AMUI-MAR systems. *Social Collaboration* is an uncommon element among the AMUI-MAR systems. Only Harrington [28] has provided this feature so that users can share their MAR information with others. *Novelty* is most associated with novel features which are integrated with AMUI-MAR systems which demark them from traditional MAR systems. The sophisticated modalities such as finger-tracking or gesture-in-the-air can influence positively the users' behaviour to explore the system.

4. Unimodal and multimodal interaction techniques

This section examines the interaction techniques used in MAR systems, distinguishing between unimodal and multimodal approaches. Unimodal interaction relies on a single input channel such as touch, speech, or gesture to communicate with the system. In contrast, multimodal interaction integrates two or more modalities, allowing for more fluid and intuitive engagement with MAR content. To contextualize the evolution toward multimodal systems, this section first investigates unimodal interaction techniques, analysing their operational characteristics and constraints. This foundational analysis provides insight into how these techniques are combined and extended within multimodal frameworks. The latter part of the section evaluates multimodal techniques and presents an impact analysis to assess how they improve upon traditional approaches, particularly in enhancing interaction flexibility and user experience in complex MAR environments.

4.1. Unimodal interaction techniques

Common unimodal interaction modalities in MAR include: (1) Touch, (2) Gesture, (3) Voice, and (4) Gaze. These individual channels have traditionally supported basic interaction tasks within MAR applications, particularly in earlier systems where the complexity of augmentation was limited. However, as MAR systems advance, incorporating dynamic environments and interactive 3D content, the limitations of unimodal interaction become more apparent. For instance, touch-based interaction may not suffice for manipulating 3D virtual elements, while voice commands alone may prove unreliable in noisy settings. Consequently, understanding the capabilities and limitations of each unimodal technique is crucial to appreciating the rationale for multimodal integration. The following subsections briefly outline each of these interaction types as implemented in MAR systems, providing the groundwork for understanding their role in multimodal configurations.

4.1.1. Tactile (touch) interaction

The introduction of smartphones or touchscreen handheld devices has led to the realization of touch-based interaction [13]. This modality is commonly used in these devices because it provides fast accessibility to embedded features and functionalities. In earlier versions of smartphones, interaction was primarily focused on touch capabilities, where the involvement of fingers was required to navigate on the touch-based screen. Consequently, interaction with MAR was traditionally based on finger usage, limiting the experience to only contact through fingers [42]. As newer versions of MAR are superseding older ones, this interaction mechanism is becoming inadequate for three-dimensional (3D) object manipulation.

Touch-based interaction has been widely employed for 2D object manipulation, including the x and y axes, while depth manipulation (z-axis) still requires further investigation [13]. Touch-based interaction on smartphones has evolved from traditional mouse and keyboard inputs to support multi-touch gestures. This includes two-finger interactions that allow users to rotate objects with near six degrees of freedom (6DOF) [13]. However, difficulties arise when mapping 2D touch points to 3D

elements for 3D object manipulation.

The challenge for interface designers of MAR systems is to implement touch capabilities that can fully leverage six degrees of freedom (6DOF), which include 3DOF for object translation and 3DOF for object rotation [13,42]. However, emerging research suggests that extending this model to seven degrees of freedom (7DOF) could significantly enhance the user experience. The 7DOF model typically introduces an additional degree of freedom such as scaling or more complex spatial manipulation enabling even more precise and intuitive control over 3D objects. This advancement is particularly relevant as MAR systems continue to evolve, pushing the boundaries of interaction beyond what was previously possible with traditional 2D and 6DOF methods.

4.1.2. Gesture interaction

The gesture interaction in a smartphone or handheld device utilizes the users' natural (gesture input) behaviour for interacting with the device. This interaction is different from tactile as discussed in subsection 4.1.1, as it involves tracking the movement of the users' hand, finger or device and then translating the gestures to a set of events or operations in MAR. Thus, the gesture interaction in MAR can be categorised into three axes: 1) Surface gesture, 2) Mid-Air gesture and 3) Motion gesture [43,44]. The surface gesture tracks the movement of users' fingers on the surface screen and maps it to an input action for interacting with the MAR system and manipulating the virtual objects [45]. Mid-Air gestures-based interaction refers to gestures which are performed outside the surface of the screen (in mid-air) to manipulate the 3D objects (Hürst &Vriens 2016). Motion gestures-based interaction employs the inertial measurement unit (IMU) to track the device movement and provide the relative interaction capability to manipulate the MAR objects [43].

The *surface gestures* work in collaboration with the touch modality as discussed in subsection 4.1.1. This mechanism tracks the movement of the fingers on the surface screen and the relative positional fingers (gestures) are then associated with a specific operation. For instance, operations such as rotating and zooming a virtual object or viewing a particular feature of the 3D object in MAR require the usage of two fingers [4]. In the same context, Wu et al. [46] implemented four types of surface gestures, including annotate, wipe, copy, and pile-n-browse to reduce task completion. Nonetheless, the screen of smartphones or handheld devices is limited, thus issues such as limited interaction space, hand occlusion of the display, and limited reach when operating single-handedly [3,10].

Mid-Air gestures or gesturing in the air refers to gestures from using the bare hands or fingers to manipulate 3D objects floating in the air. This technique tracks the movement of the user's hands or fingers on the virtual objects in free space and provides the users with a full DOF manipulation mechanism ([13]; Hürst &Vriens 2016). The tracking of the mid-air gestures is carried out using the rear camera of the mobile device to achieve the interaction. This interaction technique is commonly employed in Head-Mounted Device AR and provides an immersive experience while users interact with virtual objects. The work of Hürst & Vriens [33] has implemented basic finger-based interaction gestures such as grabbing, holding, lifting, scaling and releasing.

Motion gestures are interaction methods on handheld devices that utilise the raw data from the sensors such as inertial measurement unit (IMU) to determine the device movement in a 3D space and translate the movement to a respective feature or functionality. For example, a stationary mobile device would wake up when it experiences movement or even scrolling can be carried out by titling the handheld device. These are common instances of motion gestures which have been implemented in mainstream applications. Similarly, in MAR, motion gestures have been exploited to provide various controls such as Touch, Move and Release for manipulating MAR objects [43]. The accelerometer and gyroscope sensors in handheld devices have added new possibilities for interaction mediums. Firmanda et al., (2022) is among the few works which explored motion gestures using an instructional UI. The UI directs the user to move the smartphone so that it can scan the surface to recognise the floor and wall surfaces.

4.1.3. Speech interaction

Speech interaction also referred to as voice user interface is an interface medium in handheld devices which provides interaction and communication services to the users [5]. These handheld devices have built-in microphones and speakers to support common functionalities such as attending calls, listening to music or taking photos and recording videos. Natural User Interface (NUI) is predominantly edging in handheld devices to simplify user interaction, voice commands are alternatives that offer many features to users, including complex menus [6]. To this end, MAR applications can have access to the voice user interface to provide a natural form of interaction with both MAR objects and the system (Lugtenberg et al., 2018). Dictionary words can be added for specific MAR commands, which thus allows the users to access different features of the MAR system, including manipulation and interaction with MAR objects. This interaction comes with a default recogniser which interprets the signal to confirm if the speech is within the words applicable to the system [43]. Moreover, audio as outputs can be featured in MAR systems to provide users with auditory information. Abidin et al., (2018b) has integrated the *push-to-talk* and *near-to-talk* features in their work, which has leveraged both the speech and sound mediums in the Islamic Go system.

4.1.4. Gaze interaction

Gaze interaction or visual interface is a novel modality in the immersive virtual environment which exploits gaze as a context feature to adapt contents in the user interface [47]. The medium is widely used in immersive technologies as both input and output modalities. For instance, the tracking of eye movements is used for (1) understanding the user experience and learning experience, (2) understanding users' emotional states and visual exploration, (3) building attention models of users and (4) detecting objects and areas of interest [48]. Research into eye-gaze as an input modality in MAR is limited in nature due to its implementation complexity as it involves enabling the front camera to track and monitor the eyes' movement, while the back camera does the superimposition [49]. Nevertheless, this interaction mechanism has been investigated in MAR to activate information when the system detects eye fixation in the direction of the handheld device. There are few works which have explored the gaze modality, they have built visual attention models to extract patterns and provide content suggestions that would be interesting to the users [50–52]. In this research, Del Amo et al. [23] is the only work that has explored the visual interface. They track the performance of the maintenance workers and facilitate them to carry out their repair activities.

4.2. Review of multimodal interaction techniques in MAR

This section reviews the multimodal interaction strategies that have been proposed from the MAR systems which have been studied in Section 3. The focus is shifted to understanding the relation of the unimodal and multimodal techniques in the MAR continuum and assessing the usefulness and effectiveness of multimodal interaction techniques in the MAR systems. Notwithstanding, this evaluation should assist in modelling the multimodal interaction techniques, the integration and deployment roadmaps, the arising issues and challenges and the potential areas for improvement in the MAR environment. From the fifteen works which have been examined in Section 3, an initial categorisation is carried out to find out the common multimodal interaction techniques and then a critical analysis is performed to determine the gaps in the literature.

4.2.1. Categorisation of multimodal interaction techniques in MAR

In this subsection, the common multimodal interaction techniques are identified from the MAR systems which have been investigated in

Section 3. A relation mapping of these techniques is carried out from the unimodal ones and an initial categorisation is derived based on the various literature in the field. Table 2 presents the common multimodal interaction techniques that are employed in MAR systems.

From Table 2, a grouping of the multimodal interaction techniques from the reviewed works is carried out and it is correlated to the number of systems which have implemented this technique. The categorisation in Table 2 indicates the multimodal techniques which have been employed in most cases in MAR systems but also hints at possible gaps that can be explored further. At first glance, three out of fifteen works have opted for *tactile, surface* and *motion* modalities. These modalities are commonly implemented in MAR systems because they can be achieved with the convenience of sensors in smartphones. The initial version of MAR systems was dominated by the *tactile* and *surface* modalities as smartphones were having limited sensory capabilities. Consequently, the interaction experience was inadequate for real-time MAR systems which is also obtrusive and persuasive in nature. MAR has seen continuous evolution; the *motion* modality has been introduced to provide useful shortcuts to features and functionalities which are familiar and intuitive. The sensors such as gyroscopes, compasses and accelerometers have enabled MAR designers and developers to build robust multimodal interaction frameworks for MAR systems.

Voice input, also known as speech modality has been integrated in a wide range of applications. The conventional MAR systems were bounded with previously known modalities such as *tactile*, *surface* and *motion*. As MAR is constantly developing, the speech modality has been integrated with other common modalities to further enhance the user experience in MAR. For instance, Islamic Tourism Go [20] has leveraged this technique to provide an improved experience to visitors in finding Islamic POIs using push to talk feature. Additionally, the speech modality has been explored in tightly-spaced environments where the objects are placed close to each other. In these situations, this technique can assist the user to interact both with the physical objects and the augmented contents without impacting the real experience. Moreover, MAR systems are currently adopting novel techniques such as mid-air and gaze. These techniques are commonly employed in HMD AR systems, but they are now being exploited in mobile (smartphone) AR systems.

The *mid-air* technique is altering traditional MAR systems' concept to a more pervasive and immersive system. Previously, MAR systems were restricted to providing 2D and 3D augmented content on the surface screen of smartphones, but this technique provides the ability to interact with floating (augmented) objects in their natural forms [13]. This technique has significantly improved the perception of visualising and sensing the MAR objects. The augmented contents can be visualised and interacted in the air at any angle and the interaction can be as tangible as interacting with real objects. The *mid-air* technique works well in collaboration with other techniques such as tactile and speech modalities. Chen et al. [4] and Hürst & Vriens [33] have explored these multimodal techniques in their systems. Chen et al. [4] have exploited the *mid-air* technique in an AR game where they detect hand movements and objects. The authors have employed the leap motion technique to determine hand and finger tracking.

Table 2
Categorisation of multimodal interaction techniques.

#	Multimodal Techniques						No of Systems
	Tactile	Gesture			Speech	Gaze	
		Surface	Mid-Air	Motion			
1	Y	Y	–	Y	–	–	3
2	Y	Y	–	Y	Y	–	2
3	Y	Y	–	–	–	–	2
4	Y	–	Y	–	Y	–	2
5	Y	–	Y	–	–	–	2
6	Y	Y	Y	Y	Y	–	1
7	Y	Y	Y	Y	–	–	1
8	Y	–	–	Y	–	Y	1
9	Y	–	–	–	Y	–	1

Hürst & Vriens [33] have deployed markers to realise the potential of multimodal feedback for finger-based interaction. The system tracks the markers which are placed on the fingers and enables the user to perform select and deselect gestures with the MAR objects. The authors have used the AR library for natural feature tracking in Vuforia Software Development Kit (SDK) to achieve multimodal gestures. From another perspective, gaze modality is unpopular among the interaction mechanism in the MAR spheres. As this medium detects eye fixation, it has been assisting MAR users in dense or risky environments where the superimposition should be cognition-free, precise and accurate. Del Amo et al. [23] have adopted this interaction mechanism to help maintainers to have accurate and precise information in a hazardous environment. The system detects eye fixation to measure the performance of certain tasks and provide useful feedback to the users.

5. The adaptive multimodal interaction frameworks

The rapid advancement of MAR technology has sparked growing interest in developing frameworks that enhance user interaction. As MAR becomes more unobtrusive, it increasingly integrates into everyday life. Technological improvements such as enhanced visual effects and 3D augmentations have expanded content delivery, yet few studies have focused on frameworks that support adaptable interaction mechanisms across diverse contexts. For example, interaction modes should dynamically adjust to environmental conditions such as noise levels, lighting, or spatial constraints (e.g., indoor vs. outdoor). To address this gap, this paper introduces the concept of an adaptive multimodal interaction framework, emphasizing context-aware, flexible interaction design in MAR systems. This section analyses related frameworks, breaking down their components to better understand their structure, integration strategies, and operational requirements.

5.1. Literature review

Abidin et al. [12] proposed a framework to illustrate the adaptive multimodal interface in a location-based AR application. The authors have considered the three parameters of mobility, mobile context factor and user preference to conceptualise the framework. The term mobility refers to the users' interaction with the technology while they are on the move. They can interact with the system anytime and anywhere. The mobile context factors the time, location, social setting or security demands. Context-driven information can be available based on current users' location. The user preference focuses on users' preferred information which is derived from the users' profiles and contextual data. The system provides user-centred information based on the users' interests. These parameters are taken into consideration to build the framework. With the information provided, a graphical representation of the framework is demonstrated in Fig. 3.

The conceptual framework by Abidin et al. [12] has categorised the modalities into explicit and implicit. The explicit modalities are the user-input modalities which can be via gesture, speech or touch. The implicit modalities are associated with environmental conditions or device heterogeneity. The primitive input interactions and the raw data obtained from the explicit and implicit modalities respectively are decoded and converted into an understandable format. The multimodal adaptation module has three main functionalities. The first segment identifies the multimodal data from all the internal and external sources. The second component synthesises the data in an understandable format. The third module provides the output capability before parsing it to the AR view module.

Hashim et al. [37] proposed a framework called AR-Speech-Emotion-Marker (ARSEM) for a mobile learning system based on AR multimodal inputs. The authors considered three types of

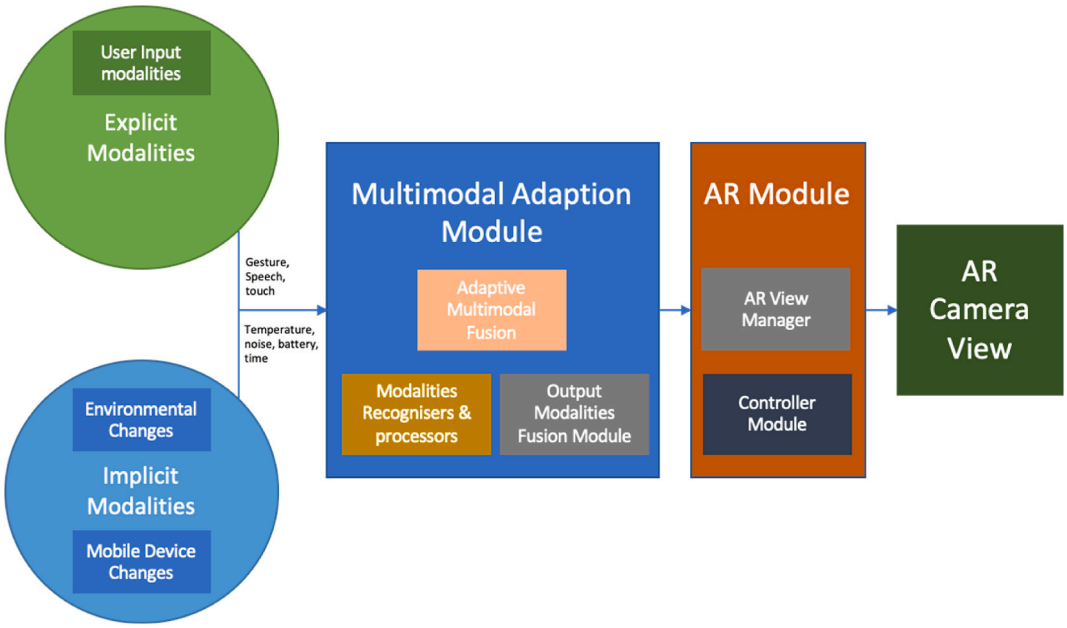

Fig. 3. Multimodal Input Adaptation Framework [12].

inputs namely emotion, image-based marker and speech. The ARSEM framework was adapted using the key components of Kolb's learning theory i.e. concrete experience, reflective observation, abstract conceptualisation and active experimentation. The four factors have assisted to identify the role of the multimodal inputs, identifying the sequence of multimodal inputs, identifying the technique to combine multimodal inputs and identifying the learning model as the basis for the learning system. An empirical study was conducted to investigate the impact of three inputs versus two inputs in learning the same emotional vocabulary. The results indicate that the three inputs have more positive impacts on the mental load, engagement, competency, challenge and learning effectiveness. Fig. 4 illustrates the framework during the active experimentation process.

The model proposed by Hashim et al. [37] has been derived from Kolb's experiential learning theory cycle. The authors have considered human emotions as an input parameter to develop the multimodal AR framework. A comparative result showed that the students' learning was better when the emotion parameter was considered in the ARSEM framework. While the same set of data was evaluated using the ARSM framework, the learning gain was slightly lower. Moreover, the additional inputs can make the learning process complicated but can generate more creativity and innovativeness in the student's learning process as they would feel more challenged. The underlying results show that the right information and the right cognitive level should have a trade-off value to achieve a good multimodal AR system. In this research, the common modalities such as tactile, gesture, speech and gaze have been identified as input. Nevertheless, the framework proposed by Hashim et al. [37] has considered emotion as an input parameter which is derived from facial expressions.

Bulagang & Baharum [35] has come up with a framework that leverages multimodal mediums in a MAR environment for higher learning education. The authors have explored the MAR framework proposed by Jamali et al. [53] to formulate a framework which would be centred to enhance the learning through MAR. Jamali et al. [53] have designed a framework that integrates various multimodal and hybrid interfaces to enhance student-centred learning through MAR. Moreover, they have provided limited information on the types of interfaces employed, and the adaptive components including the fusion mechanism and AR perspective view. The mediums such as voice and 3D interaction have been mentioned to create a multimodality experience. To the best of our knowledge, the framework of Bulagang & Baharum [35], called Net mAR has been modelled in Fig. 5.

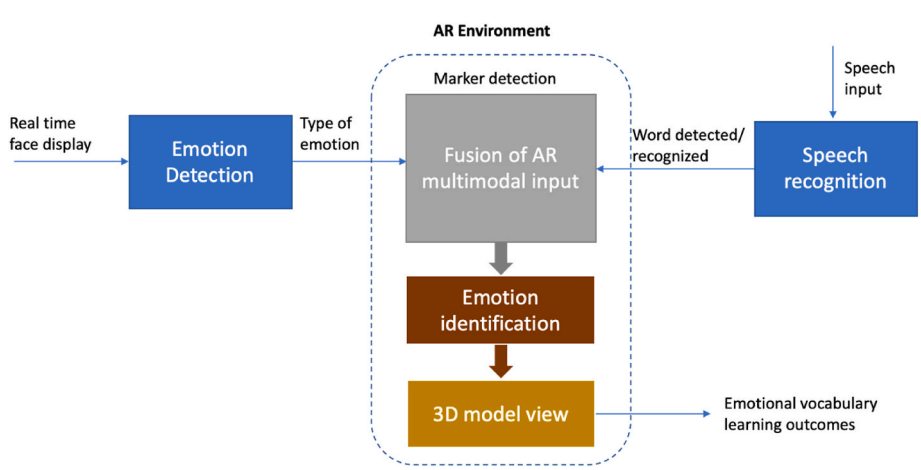

Fig. 4. Experiential learning multimodal AR framework.

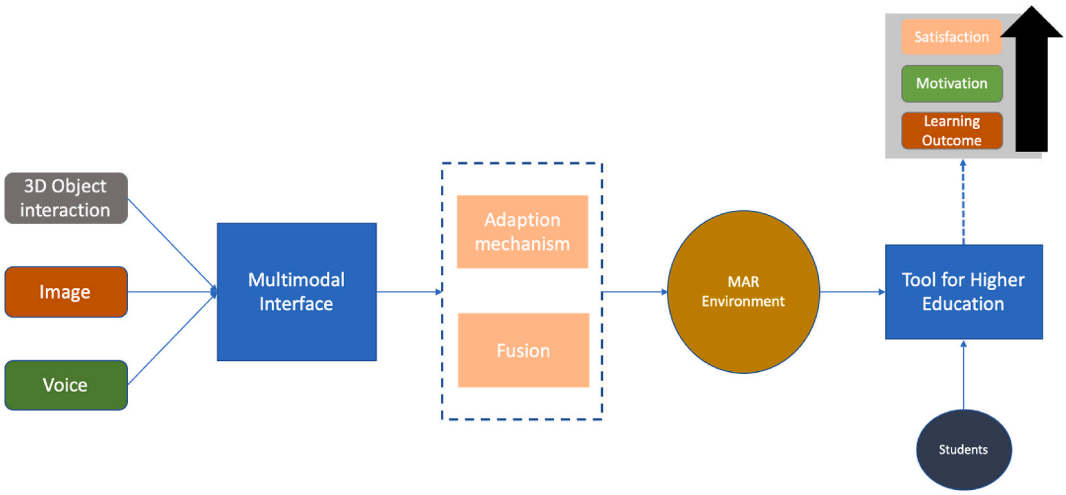

Fig. 5. Net mAR Framework.

The Net mAR framework has been formulated to assist students to learn network devices through multimodal interfaces through various mediums. The mediums such as 3D object interaction and voice can be beneficial in a learning-centred context as they will provide students with various interfaces to interact with the network devices. The purpose of the framework is to assist students to learn network devices naturally. However, the framework has limited information on the adaptation mechanism and fusion which has been highlighted using dash lines. These elements are crucial to delimit the various modalities and adjust the interaction mechanism according to the environmental condition.

Krings et al. [36] proposed a context-aware AR application framework in another study. The work is aligned with the agenda of this research as it involves context-aware and adaptation modules which take care of adjusting the AR application according to the user, their situation and their needs. The framework has three main components: context monitoring, adaptation and decision-making. The framework is illustrated in Fig. 6. The context information is measured using the smartphone sensors such as the camera and gyroscope. The decision-making module monitors the active conditions and actions from the context monitoring and adaptation components. This component processes the information parsed through the context monitoring module and then decides on whether to pass it to the adaptation layer. The adaptation module is responsible to react to the context change by adjusting the interaction modalities (surface interaction and sound) and augmented contents.

A prototype MAR application for printer maintenance has been developed using the context-aware multimodal framework. The application guides users through the process of changing the ink cartridges of a printer. Depending on certain conditions, the lighting of the screen can be adjusted to provide an enhanced user experience or the voice interface/commands can be activated if the user is intervening on the printer. Yet, there is a limited amount of information on different interaction modalities such as gesture and gaze in this framework which could further assist the users to carry out their tasks effectively.

5.2. *Synthesized conceptual framework for AMUI-MAR systems*

Drawing from the analysis of existing systems and frameworks reviewed in the Section 5.1, this section presents a unified diagram that consolidates the recurring architectural elements and interaction patterns across prior AMUI-MAR implementations. The objective is not to propose a novel framework, but rather to abstract and fuse the shared structural and functional components identified across studies. This synthesized model offers a generalized view that can support comparative understanding and serve as a reference point for future evaluations. The gap analysis conducted earlier highlighted key issues and contextual parameters shaping AMUI design, enabling the consolidation of common layers and modules. Fig. 7 visualizes this synthesis, capturing four primary layers observed across the literature: (1) Context Monitoring, (2) Adaptation Decision Module, (3) Multimodal Input Processing, and (4) Interaction Output. This layered structure encapsulates how adaptive multimodal interaction is typically orchestrated in MAR environments.

The **Context Monitoring Layer** serves as the foundation of adaptive multimodal interaction in MAR systems by continuously capturing real-

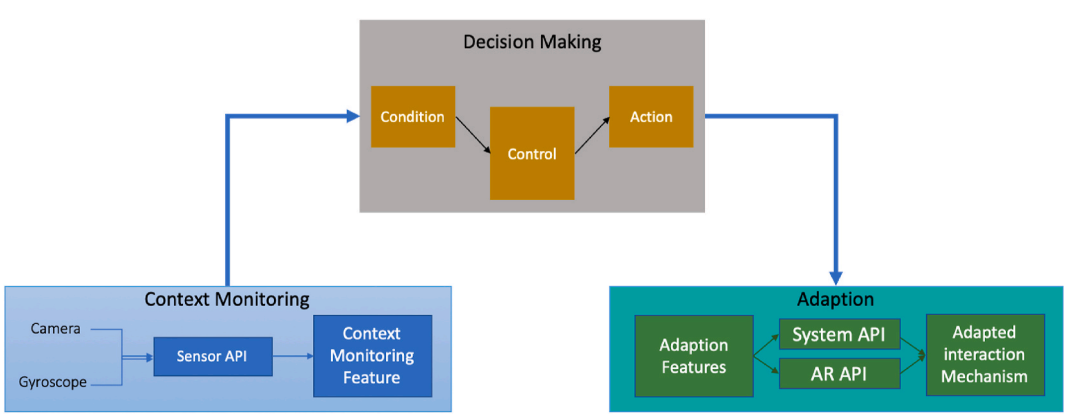

Fig. 6. Context-aware multimodal framework.

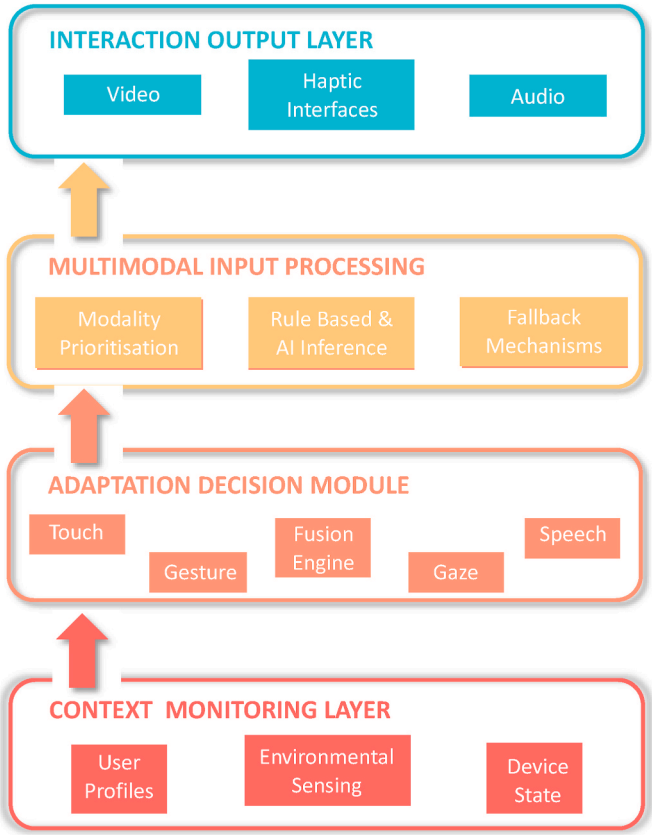

Fig. 7. Synthesized conceptual framework for AMUI-MAR systems.

time contextual information. This includes environmental parameters such as ambient noise levels and lighting conditions, as well as device-specific data like orientation, battery status, and connectivity. Additionally, this layer gathers user-centric information, including location, historical usage patterns, profiles, and preferences. Collectively, this contextual input enables the system to make informed decisions about modality selection and adaptation, ensuring that interactions remain relevant and efficient across varying situational conditions.

The **Multimodal Input Processing Layer** handles both explicit and implicit user inputs. Explicit modalities refer to direct user interactions such as touch, speech, and gestures, while implicit modalities involve passive input from environmental factors or device states, like motion or brightness. This layer also integrates a **fusion engine**, which is responsible for processing concurrent inputs across multiple modalities. By combining these inputs, the engine generates a unified representation that is passed to the adaptation module, facilitating coordinated and responsive multimodal interactions.

At the core of the framework lies the **Adaptation Decision Module**, which governs how interaction modalities are prioritized, activated, or suppressed in response to the contextual data collected. This module may operate on predefined rule-based logic or intelligent algorithms that evaluate the relevance and reliability of each modality. For example, it can switch from voice to gesture interaction in noisy environments or recommend disabling certain modalities due to system constraints. It also manages fallback mechanisms to ensure robustness, thereby maintaining continuity in user interaction even when specific modalities fail or underperform.

The final component, the **Interaction Output Layer**, delivers the selected interaction modality and augmented content to the user. This layer ensures that the output interface aligns with the contextual needs and user expectations, completing the adaptive loop. By dynamically adjusting the modality and interface presentation, this layer enhances both the usability and engagement of MAR applications, promoting seamless, intuitive, and responsive experiences.

5.3. Review of adaptive multimodal interaction frameworks for MAR

In this subsection, a review of the adaptive multimodal user interface (AMUI) frameworks for the MAR system is carried out. A comprehensive narrative of the AMUI frameworks for MAR is provided in Section 5.1. The various features, components, layers and flows are described to have a better understanding of the core operation and functioning. An analysis of the frameworks has been carried out in Table 3. The analysis provides insightful information on the input and output modalities, the intrinsic and extrinsic parameters, the main processing components (adaptive, fusion and identifier) and the AR processing view. As this domain is still in its infancy stage, very few AMUI frameworks for MAR are available.

Table 3 provides a review of the AMUI frameworks which have been studied in Section 5.1. Abidin et al. [12] provide a comprehensive AMUI framework that aims to provide a multimodality MAR experience by leveraging various input and output modalities. Moreover, the framework has an adaptive mechanism which considers the intrinsic and extrinsic parameters and then adjusts the interactivity accordingly. Yet, the work of Abidin et al. [20] has not explored complex gestures such as mid-air or gaze which could have enhanced the user experience significantly. The ARSEM framework proposed by Hashim et al. [37] has taken a unique approach compared to its predecessors. The authors have taken the user's emotions as an intrinsic parameter to provide students with more appropriate augmented content. However, this work has not investigated the adaptive mechanism in its natural format which could provide the students with different output modalities to examine the augmented objects. The Net mAR framework [35] has been devised as a learning platform. This work has not properly delineated various core features of an AMUI framework for MAR. The authors have mentioned the three mediums as follows: 3D object interaction, image and voice, but limited information is available on the fusion module or adaptive mechanism. Krings et al. [36] proposed a context-aware AR application framework. The framework covers context-aware capabilities which can read data from sensors and then adapt the AR application according to the user or situation. Nevertheless, the interaction mechanism is limited to surface gestures only.

5.4. The impact of AMUI-MAR frameworks and systems on UX and UE

The Human-Computer Interaction (HCI) factors related to the User Experience (UX) and User Engagement (UE) are still an unexplored area in the AMUI-MAR domain. The previous sections have provided an in-depth analysis of all the functional and technical aspects of the AMUI framework for the MAR system. The AMUI framework is provisioned to elevate the interaction (user) of MAR systems and provides an enhanced UX and UE. Yet, these factors have been thoroughly tested in the traditional MAR system. Few research studies have investigated the UX and UE factors in the AMUI-MAR system. Such systems are built with multiple interfaces and they can adapt accordingly to the current environment or the users' needs or some pre-empt values. In the same context, there is a need to understand the evolution of the traditional MAR system to the AMUI-MAR system and to have a mapping difference of the UX/UE factors between the traditional MAR system and the AMUI-MAR system. This section provides a qualitative assessment of the UX and UE factors for the AMUI-MAR system in various application domains.

5.5. Impact of AMUI-MAR frameworks on UX and UE

The work of Hashim et al. [37] conducted an empirical study to investigate the impact of the different multimodal inputs on mobile AR learning environment. The authors initially carried out an impact assessment on the students' learning with two inputs and then they

Table 3
Review of AMUI frameworks for MAR.

#	AMUI Frameworks	Modalities		Parameters		Main Components
		Input	*Output*	*Intrinsic*	*Extrinsic*	
1	Abidin et al. [12]	Gesture, Speech & Touch	3D object & Voice	Users' profiles & locations	Noise, Time, Device settings & Environmental conditions	Modality recogniser, Adaptive mechanism & Output module.
2	ARSEM [37]	Facial, Touch & Speech	3D model	User emotions	–	Modality fusion & Emotion identification.
3	Net MAR [35]	Touch & Speech	3D object & Voice	Students Profiles	–	Multimodal fusion
4	Krings et al. [36]	Touch, Speech & Gesture	3D object, voice	Users' profiles	Context information	Context monitoring, Decision making & adaptation mechanism.

evaluated the framework with three inputs. The results showed that the extra input in the ARSEM framework has provided positive impacts on the mental load, engagement, competency, challenge and learning effectiveness. Yet, the authors stressed that the addition of more inputs has a propensity to make the learning process complicated. In this situation, the students accepted the challenge of using the three inputs. The results showed that cognitive load was lower compared to the results obtained from the evaluation carried out with the two inputs. As stated by the authors, a small mental load will keep the students engaged else they will feel the task is too easy and thus they will disengage with the system. A proper trade-off of the cognitive load (complexity) and the number of interaction mediums should be devised when designing such systems.

Abidin et al. [12] evaluated the framework proposed in their previous work of Abidin et al. [20] through a prototype implementation of an AMUI-MAR system in a tourist site. Three interaction mediums have been implemented in the system as follows: speech, motion and touch. Additionally, the environmental and user contexts have been taken into consideration to provide an adaptive interaction mechanism. The aim of the evaluation was two-folded. The authors initially assessed the users' satisfaction and they evaluated the usability of the application at the second stage. An average score of 75 % is reported from the evaluation results, which the authors have rated as a good score. Moreover, the users' feedback and comments were mainly focused on improving the AR information and the functionalities of the AR system. No issue was reported regarding the multimodal interaction mechanism.

Krings et al. [36] considered the UX aspects primarily when designing the AARCon framework. The framework considers the contextual and user parameters to make the application more ergonomic and easier to work with. The framework is depicted in Fig. 6. A prototype implementation of a MAR system using the framework has been carried out. The authors tested the prototype for the maintenance of printers. During a printer maintenance task, two hands are often required to manipulate physically with the printer e.g. changing the toner. Thus, the system activates the voice interface to continuously assist the users during the process. The system is equipped with several interaction mediums and adjusts the AR information according to the users' field of view. The authors stated that the system has enhanced the understanding of users when they are working with the printer.

A brief comparative analysis of the frameworks in relation to the UX and UE is summed in Table 4. The works of Ramtohul & Khedo [38] and Ramtohul & Khedo [39] have provided sufficient information on the design characteristics and the elemental UX and UE framework in the MAR domain. The AMUI-MAR frameworks have been assessed using the most suitable UX and UE factors from the works of Ramtohul & Khedo [38] and Ramtohul & Khedo [39].

The analysis from Table 4 provided insights into the UX and UE factors of the AMUI-MAR frameworks. Both the UX and UE factors are important while designing an AMUI-MAR framework. This framework is designed in such a way as to provide an adaptive multimodality experience to the users, thus enhancing the experience and engaging the user in the process. From the assessment in Table 4, it can be perceived that AMUI-MAR frameworks have an impactful UX and UE rather than traditional MAR frameworks. The conventional frameworks focused mainly on providing a useable system rather than concentrating on a user-centric system. The AMUI-MAR frameworks encompass various futuristic UX and UE elements such as connectedness, persuasive, mobility, recognition and familiarity. As a result, such frameworks have enabled researchers and designers to focalise their design on certain UX and UE aspects, which should have a greater influence on the UX and UE.

Table 4
Comparative UX and UE analysis of AMUI-MAR frameworks.

Frameworks	Featured UX Characteristics	Featured UE Characteristics	Comments
ARSEM	Usefulness, Effectiveness, Appealing & Consistency	Novelty, Learning, Emotion & Challenge	Field study shows that the framework has weighted on both UX and UE elements. Yet, more UE features such as collaboration and immersion could have been added to supplement the engagement level.
Abidin et al., (2018a)	Accessibility, Mobility, Recognition & Familiarity	Personalisation, Novelty, Persuasive & Connectedness	Framework provided a robust UE in terms of adapting to the users' or environmental context. Elements such as tangibility and context-layering could have reinforced the existing UX.
Krings et al., (2020)	Accessibility, Familiarity, Feedback & Attention	Adaptive	Framework has been designed for a specific application field, thus important UX and UE elements are overlooked.

6. Research challenges and future directions

Based on the findings of this research, several key gaps and areas for further development have been identified in the field of AMUI-MAR. As AMUI-MAR is still in its infancy stage, it is important to note that the reviewed systems were limited in number. Therefore, future research should aim to include a more diverse range of AMUI-MAR systems for a comprehensive understanding. The research gaps are summarised below.

6.1. Real-time multimodal fusion and synchronisation

This research has focused primarily on decomposing the various interaction components of AMUI-MAR systems including the interaction algorithms. The design and development of such systems lay on a robust interaction mechanism that should be taken care of. The interaction

Table 5
Evaluation of AMUI-MAR systems.

AMUI-MAR Systems	*UX and UE Factors*								
	Familiarity	*Direct Manipulation*	*Intuitiveness*	*Personalisation*	*Liveliness*	*Sense of Presence*	*Multi-Interaction*	*Social Collaboration*	*Novelty*
Islamic Tourism Go (Abidin et al., 2018b)	Y	Y	Y	Y	Y	Y	Y	–	Y
Chen et al., (2017)	Y	–	–	–	–	–	Y	–	–
Srivastava & Yammiyavar [21]	Y	–	–	–	–	–	Y	–	–
Vannotator (Abrami et al., 2019)	Y	–	Y	–	–	–	Y	–	–
Del Amo et al. [23]	Y	–	–	Y	–	–	Y	–	Y
Hellmuth et al., (2021)	Y	Y	–	–	Y	Y	Y	–	Y
TourMar [25]	Y	–	–	Y	–	–	Y	–	–
Firmanda et al., (2022)	Y	–	Y	–	–	–	Y	–	Y
Harrington [28]	Y	Y	Y	–	Y	Y	Y	Y	Y
AR Sightseeing (Matviienko et al., 2022)	Y	–	Y	–	Y	Y	Y	–	–
Gül [30]	Y	–	–	–	–	–	Y	–	–
Schraffenberger & Heide [31]	Y	–	–	–	–	–	Y	–	–
Ismail et al., (2019)	Y	Y	Y	–	Y	–	Y	–	Y
Hürst & Vriens [33]	Y	Y	Y	–	Y	Y	Y	–	Y
Lugtenberg et al., (2018)	Y	Y	Y	–	–	Y	Y	–	Y

algorithm provides a seamless, intuitive and natural way for users to interact with augmented objects. However, further research is warranted to explore the fusion of multimodal input in AMUI-MAR frameworks. Investigating how different modalities can be seamlessly integrated and synchronized to create more immersive and interactive experiences holds great potential for future developments.

6.2. Framework generalization and contextual integration

The AMUI-MAR frameworks have shed light on the vital component of such systems and how they are connected. As this topic is still emerging, a limited amount of information is available. Similar frameworks can be investigated to have a broader understanding of retrieving more contextual data and integrating it into the model.

6.3. Object level vs. system-level interaction

The main aim of this research was to explore the adaptive multimodal user interfaces in MAR systems and their implications. In this study, more attention has been given to the interaction with MAR objects rather than the interaction with the system. Interaction with the MAR objects involves manipulating the augmented object while interacting with the system encompasses accessing all the functional features of the application. A proper demarcation of the interaction with MAR objects and interaction with MAR systems should be carried out.

6.4. Enriching contextual adaptation through user-centric factors

The implicit and explicit modalities play an important role in providing an adaptive contextual mechanism to the users. In this research, the work of Abidin et al. [12] has emphasized these terms to stem the term *adaptive* in their proposed framework. However, the implicit and explicit modalities are limited to user input, environmental information and mobile device changes. More factors such as previous experience, the pattern of movement and users' cognitive and emotional factors can be considered as future avenues for research.

6.5. Implementing and validating adaptive fusion models

The adaptive multimodal fusion model is a crucial component in the AMUI-MAR context. The role is two-folded. It deciphers the interaction modalities from implicit data and then combines the various interaction mediums to provide a multimodality experience. Moreover, the explicit data adjust the experience to an adaptive multimodality one. This research has theoretically described this component, thus it might be subject to enhancement during the implementation phase.

6.6. Clarifying overlaps in tactile and surface interaction

In this research, tactile and surface interaction have been treated independently. These modalities can be conflicting as they share similar characteristics and operate dependently with each other. Research has characterised tactile and surface interaction as the same. To the best of our knowledge, these two modalities have been differentiated and a classification has been carried out. However, further research should be carried out to have the best differentiation and classification of these two modalities.

6.7. Exploring tangible and embodied interaction modalities

In the quest to improve users' interaction in MAR systems, new types of interaction are emerging. Tangible interaction is becoming appealing in MAR as it involves a multifaceted mechanism which makes the user feel that they are physically engaging with the MAR objects. This works in collaboration with mid-air, audio and vibratory modalities which invokes a perception that users can sense the thickness of MAR objects. This opens new avenues for research in this field.

6.8. Leveraging AI/ML for dynamic modality adaptation

Although this study has focused on defining the interaction and fusion mechanisms in AMUI-MAR systems, the role of Artificial Intelligence (AI) and Machine Learning (ML) remains underexplored. These technologies can be instrumental in dynamically predicting the most effective modality (or combination of modalities) based on real-time user behaviour, context, and environmental cues. Future research should investigate how AI/ML models, particularly deep learning and reinforcement learning approaches can enhance the adaptivity of multimodal interfaces, improve decision-making accuracy, and reduce user fatigue in complex MAR scenarios.

7. Conclusion

The systematic review of adaptive multimodal user interface techniques (AMUI) in mobile augmented reality (MAR) conducted in this study reveals valuable insights into enhancing user experience (UX) and user engagement (UE). The traditional interaction techniques are still being employed in MAR which limits the user to have a natural

interaction with the MAR objects, thus impacting negatively on the UX and UE. The novel types of MAR have invigorated new forms of interaction mechanisms. The AMUI techniques have been devised to improve the UX and UE in MAR by employing a combination of various interaction modalities so that users can interact naturally with the MAR objects. This research adopted a comprehensive approach to explore AMUI techniques in MAR, encompassing: (1) understanding adaptive and multimodal terminologies, (2) differentiating and categorizing unimodal and multimodal interaction modalities, (3) examining key components of AMUI-MAR frameworks, and (4) evaluating the implications of UX and UE on AMUI-MAR systems. This research revealed five significant findings. They are listed as follows:

- The first outcome reveals the widespread usage of the leap motion algorithm in tracking users' hand or finger movements in AMUI-MAR systems.
- The second finding highlights the 7DOF algorithm as a promising successor to the 6DOF algorithm, enabling real-time rotation and translation.
- The third outcome underscores the prevalent use of tactile, surface, and motion interaction modalities in AMUI-MAR systems.
- The fourth finding establishes a strong association between mid-air gestures and speech/sound modality in AMUI-MAR interactions.
- Lastly, the importance of investigating the fusion of multimodal input is emphasized, as it represents a key component of the AMUI-MAR framework.

This systematic review of AMUI in MAR has shed light on the potential of these techniques to significantly enhance UX and UE in MAR. By deepening the understanding of AMUI techniques and their impact on UX and UE in MAR, it can pave the way for more compelling and user-centred MAR interactions. The advancements in AMUI-MAR frameworks have the potential to revolutionize how we perceive and interact with digital content, offering seamless integration of virtual and physical worlds. Continued exploration and refinement of AMUI techniques will undoubtedly shape the future of augmented reality, opening up new avenues for creativity, productivity, and immersive experiences.

CRediT authorship contribution statement

Arvind Ramtohul: Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Kavi Kumar Khedo:** Writing – review & editing, Validation, Supervision.

Funding statement

The authors did not receive support from any organization for the submitted work.

Declaration of competing interest

The author(s) declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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